MILITARY SPECIFICATION

INDICATOR, LIQUID OXYGEN QUANTITY

This specification is approved for use by the Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for liquid oxygen quantity indicators. The indicators are of the capacitance type and indicate from 0 to 25 liters of liquid oxygen.

1.2 Classification. The indicators shall be of the following types, as specified (see 6.2):

GMU-28/A	Cover glass coated with antireflection coating in accordance with MIL-C-675
GMU-28A/A	Cover glass coated with high efficiency antireflection coating as defined herein.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood Cleated-Plywood
PPP-B-636	Boxes, Fiberboard

MILITARY

MIL-P-116	Preservation, Method of
MIL-C-675	Coating of Glass Optical Elements (Anti-Reflection)
MIL-I-25645	Indicator, Liquid Oxygen Quantity, Capacitance Type, General Specification for
MIL-L-27160	Lighting, Instrument, Integral, White, General Specification for

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, Texas 78241 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

	FED-STD-595	Colors

MILITARY

*	MIL-STD-129	Marking for shipment and storage
	MIL-STD-147	Palletized Unit Loads
*	MIL-STD-781	Reliability Tests, Exponential Distribution
*	MIL-STD-794	Part and Equipment Procedures for Packaging and Packing of
*	MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking and Waterproofing with Appropriate Test Methods
	MS33558	Numeral and Letter, Aircraft Instrument Dial, Standard Form of

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.1.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

* AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951	Standard Practice for Commercial Packaging

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

* 2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

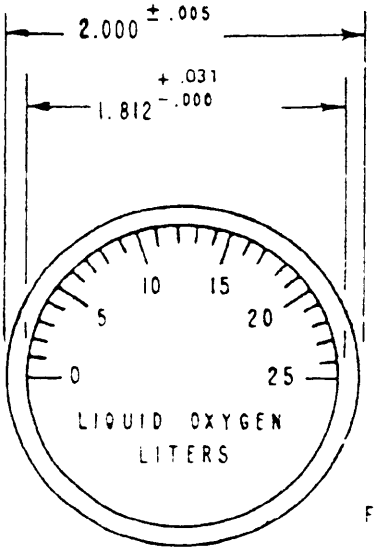
3. REQUIREMENTS

3.1 First Article. When specified, a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.2 General specification. The requirements of MIL-I-25645 shall apply as requirements of this specification with the exceptions and additions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Configuration. The indicator shall consist of a single unit containing all the equipment necessary to meet the performance requirements specified herein and in MIL-I-25645.

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NUMERALS & LETTERS SHALL CONFORM TO MS 33558

MARKING	HEIGHT OR LENGTH ±.010	WIDTH OF LINE OR GRADUATION ±.005	FINISH COLOR PER FED-STD-595
Numerals	.140	.025	37875
Liquid Oxygen	.140	.020	
Major Grad	.188	.031	
Minor Grad	.125	.015	
Liters	.100	.015	
Background of Dial			37038

UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES.

FIGURE 1. Dial

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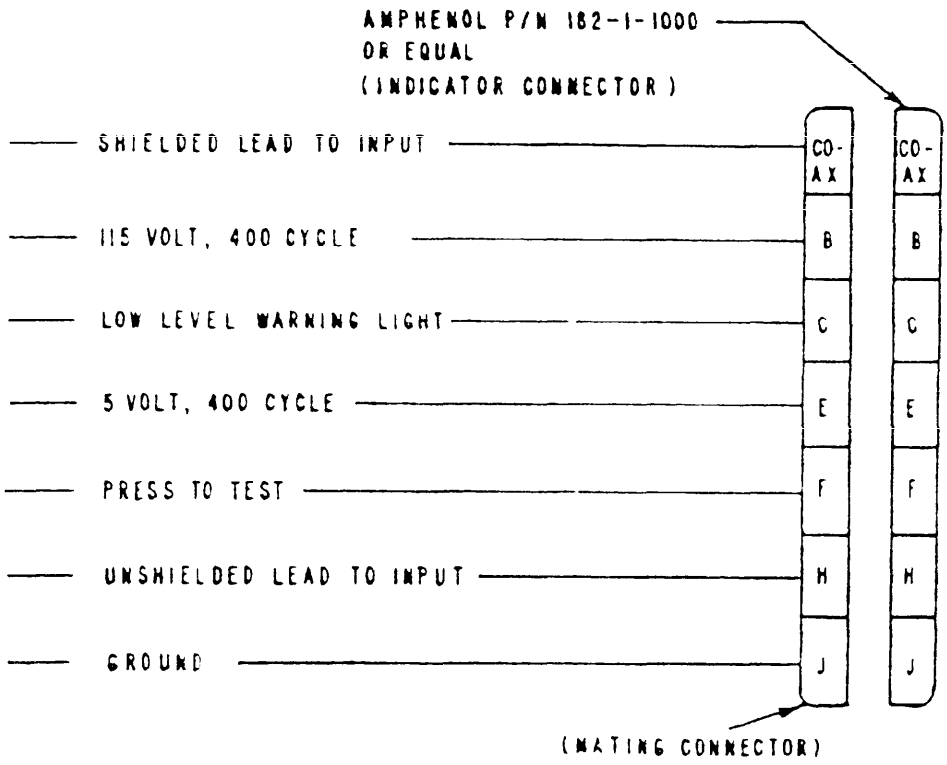


FIGURE 2. Connector Wiring

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3.4 Case. The length of the case shall not exceed 5.0 inches.

3.5 Dial. The dial for the indicator shall conform to figure 1.

3.6 Cover glass coating (GMU-28A/A). The indicator cover glass for the GMU-28A/A indicator shall be provided with a reflection-reducing coating which meets the requirements of MIL-C-675 except that the following reflectance tolerances shall apply:

Angle of Incidence	Wavelength (Millimicrons)	Percent Reflectance
0°	450 - 675	0.6 maximum
0°	425 - 700	0.5 average
30°	450 - 625	1.0 maximum
30°	425 - 700	0.5 average

3.7 Electrical connector. The electrical connector shall be in accordance with figure 2. The indicator shall be designed internally so that it will function properly when the mating electrical connector is wired as shown on figure 2. The coaxial shield shall be grounded within the indicator.

3.8 Capacitance. For design purposes, the capacitance for zero reading shall be 303.5 micromicrofarad (uuf).

3.9 Adjustments. The zero calibration adjustment for the indicator shall be capable of adjusting the pointer to zero indication over an input capacitance range of at least 295.0 uuf to 330.0 uuf. With the indicator adjusted to zero indication at 295.0 uuf input capacitance, the indicator shall be capable of calibration to full scale indication at 437.5 uuf input capacitance by means of the full calibration adjustment. With the indicator adjusted to zero indication at 330.0 uuf input capacitance, the indicator shall be capable of calibration to full scale indication at 475.0 uuf input capacitance by means of the full calibration adjustment.

3.10 Weight. The indicator shall weigh not more than 1.5 pounds.

3.11 Accuracy. When the indicator is tested as specified in MIL-I-25645, the accuracy of the indicator shall be in accordance with table I. The tolerance for room-temperature readings is the allowable deviation from the nominal calculated values. All other tolerances are the allowable deviations from the actual room-temperature readings.

3.12 Integral lighting. The indicator shall have integral lighting conforming to MIL-L-27160.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other

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facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

TABLE I. Tolerances for Capacitance Readings

TEST	TOLERANCE PLUS OF MINUS
Accuracy - room temperature	1.8 uuf
Low temperature operation	2.5 uuf
High temperature operation	2.5 uuf
Vibration	1.4 uuf
Power supply variation	1.4 uuf
Low temperature storage and operation	2.5 uuf
High temperature storage and operation	2.5 uuf
Salt spray	1.4 uuf
Cycling	1.4 uuf

4.2 Sampling, inspecting, and testing procedures. Sampling, inspecting, and testing procedures shall be in accordance with MIL-I-25645.

* 4.3 First Article Inspection.

* 4.3.1 First Article Test - The first article test shall consist of all tests described under tests methods of MIL-I-25645 including reliability and longevity tests which shall be conducted in accordance with MIL-STD-781. The GMU-28A/A indicator shall be subjected to the additional test described in 4.4.1 of this specification.

4.4 Test methods

4.4.1 Cover glass reflectance (GMU-28A/A). The reflectance of the GMU-28A/A indicator cover glass shall be measured and shall be within the tolerances specified in 3.6.

* 5. PACKAGING.

* 5.1 Preservation and packaging. Preservation and packaging shall be Level A, C or Commercial/Industrial as specified (see 6.2).

5.1.1 Level A. The indicator shall be preserved and packaged in accordance with method 1C of MIL-P-116 and placed in a container conforming to PPP-B-636.

5.1.2 Level C. The Level C preservation for the indicator shall conform to the MIL-STD-794 requirements for this level.

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5.1.3 Commercial. The Commercial preservation of indicator shall be in accordance with the requirements of ASTM D 3951.

5.2 Packing.

5.2.1 Level A. The indicator preserved and packaged as specified in 5.1 shall be packed in a container conforming to PPP-B-601, overseas type. The closure shall be in accordance with the appendix of the container specification.

5.2.2 Level B. The indicator shall be preserved and packaged as specified in 5.1.1 shall be packed in fiberboard containers conforming to PPP-B-636 Weather Resistant. The closure shall be in accordance with the appendix of the container specification.

5.2.3 Level C. The indicator shall be preserved and packaged to requirements of MIL-STD-794 for this level.

5.2.4 Commercial. Commercial/industrial packing of indicator shall be in accordance with the requirements of ASTM D 3951.

5.2.5 Palletized Loads. When specified (See 6.2) unitized loads, commensurate with the level of packing specified in the contract or order, shall be palletized in accordance with MIL-STD-147. Unitized loads shall be uniform in size and quantities to the greatest extent possible. If the container is of a size which does not conform to any of the pallet patterns specified in MIL-STD-147, the pallet pattern used shall first be approved by the contracting officer.

5.3 Physical protection. Cushioning, blocking, bracing and waterproofing shall be in accordance with MIL-STD-1186. Waterproofing requirements for cushioning material and container shall be waived for domestic shipments. The drop tests of MIL-STD-1186 shall be waived when the items are preserved and packaged for immediate use or when the drop tests, MIL-P-116 are applicable.

5.4 Marking.

5.4.1 Level A, B & C. In addition to any special or other identification marking required by the contract (see 6.2) each unit pack intermediate and exterior container and unitized load shall be marked in accordance with MIL-STD-129.

5.4.2 Commercial/industrial marking shall be in accordance with the requirements of ASTM D 3951.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use in indicating the quantity of liquid oxygen remaining in the converters in C-5A, C-130E, and C-141 airplanes.

* 6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification
b. Type of indicator required (see 1.2)
* c. Selection of applicable levels of preservation, packaging, packing, marking and commercial/industrial.
d. When palletized loads are required (see 5.2.5).

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e. When a repeater indicator is required, when possible

6.3 First Article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory. The first article should consist of single unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.4 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Review activities:
Air Force - 11
DLA - GS
Navy AS
International interest (see MIL-I-25645)

Preparing Activity:
Air Force - 82

Project 6680-F208

* U.S. GOVERNMENT PRINTING OFFICE: 1989—505-038/21758

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-27544C	Indicator, Liquid Oxygen Quantity

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)* _____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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